Continuity and the Intermediate Value Theorem

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.02

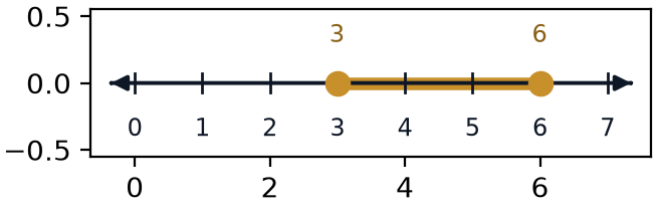
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A rectangular field is to be enclosed by a fence. One side of the field is adjacent to a river and does not require fencing. The farmer has 120 meters of fencing material. Let the dimensions of the field be x meters perpendicular to the river and y meters parallel to the river. The area of the field is given by $A = xy$. Part A: Express the area A as a function of x. Part B: Determine the dimensions x and y that maximize the area of the field. Provide the maximum possible area.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the function $f(x) = 4$. What is the value of the definite integral $\int_3^6 f(x)\,dx$?



3 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider a region in the first quadrant bounded by the curve $y = 9 - x^2$, the x-axis, and the y-axis. A rectangle is inscribed within this region such that one vertex is at the origin, two sides lie along the x and y axes, and the fourth vertex lies on the curve $y = 9 - x^2$. If the area of this inscribed rectangle is maximized, determine the value of the definite integral that represents the volume of a solid formed by revolving the region bounded by the curve $y = 9 - x^2$, the x-axis, and the y-axis about the y-axis.

4 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A particle's velocity is given by $v(t) = 3t^2 - 12t + 9$ for $t \geq 0$. The position of the particle at time $t = 0$ is $s(0) = 5$. \\ Consider a second particle whose position $p(t)$ is described by the integral $p(t) = \int_0^t v(u)\,du$. \\ Find the net displacement of the first particle from $t = 1$ to $t = 3$.

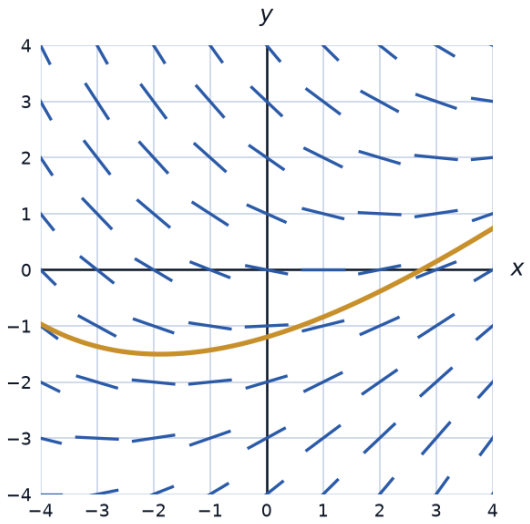
5 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Consider the function $f(x)$ defined by $f(x) = \frac{x^2 - 9}{x - 3}$ for $xeq3$. Let $g(x)$ be a function such that $g(x) = \{f(x)$, if $x < c$; k, if $x \geq c$. If $g(x)$ is continuous at $x = c$ and the limit of $g(x)$ as x approaches c from the left is 5, determine the value of k.

© ClipSAT — clipsat.org | All Rights Reserved

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The slope field for a differential equation is shown above. The differential equation is of the form $\frac{dy}{dx} = ky$ for some constant k. If the solution curve passes through the point $(0, 1)$ and $(1, 2)$, find the value of k.



7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Consider the function $y = g(x)$ and its inverse $x = h(y)$. The tangent to the graph of $g(x)$ at $x = a$ has a gradient of $\frac{1}{3}$. If $g(a) = 5$, what is the gradient of the tangent to the graph of $h(y)$ at $y = 5$?

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A differentiable function $f(x)$ is such that its derivative $f'(x)$ is defined by $f'(x) = 3x^2 - 12x + 9$. The function $f(x)$ has a local maximum at $x = 1$ and a local minimum at $x = 3$. The graph of $y = f(x)$ passes through the point $(0, 5)$. Consider the tangent line to the graph of $y = f(x)$ at the point where $x = 0$. What is the y-intercept of this tangent line?

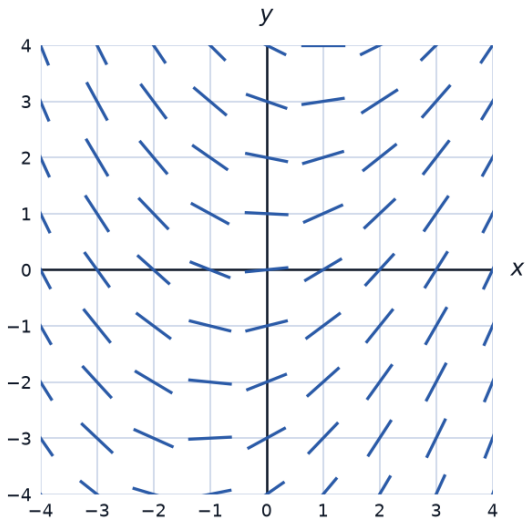
9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A differentiable function $y = f(x)$ satisfies the differential equation $\frac{dy}{dx} = 3x - 2$. If $f(1) = 5$, what is the value of $f(3)$?

© ClipSAT — clipsat.org | All Rights Reserved

10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Level 3 · AdvancedA differential equation $\frac{dy}{dx} = (y-1)^2\cos(x)$ has initial condition $y(0) = 2$. Find the particular solution $y = f(x)$.



11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Level 1 · FoundationsOn a Rate In / Rate Out FRQ, if rain falls into a cistern at rate $R(t)$ and is pumped out at rate $P(t)$, when is the amount of water in the cistern at a relative maximum?

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Level 1 · FoundationsHow many total questions are on the AP Calculus AB exam, and what percentage of the composite score comes from Free-Response questions?

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

[AP Calculus AB Free-Response Questions (FRQs) · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Level 1 · FoundationsOn an AP Calculus BC exam question, a student writes $\int_0^5 (R(t) - D(t))\,dt = 42.18$ as their final response to a question asking for the total amount of water in a tank at $t = 5$, given initial volume 100 gallons. Identify the missing component in the student's solution.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

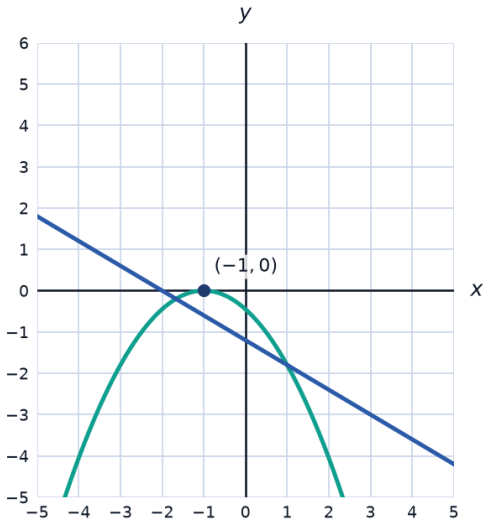
15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 2.01 The Definition of the Derivative**

Level 2 · MediumWhy does College Board recommend storing intermediate values in graphing calculator memory variables rather than writing down a 3-decimal rounded number and reusing it?

© ClipSAT — clipsat.org | All Rights Reserved

16 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives**

Level 3 · AdvancedA student writes: 'It is increasing because it's derivative is positive.' Why will this response lose credit on an AP exam?



17 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 The Chain Rule**

Level 3 · Advanced / AP FRQLet $f(x) = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$. Identify the closed-form elementary function that this series represents, and determine its radius of convergence.

18 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions**

Level 2 · MediumOn Section II Part A (Calculator Active), a student is asked for the acceleration vector at $t = 3$ for a particle whose velocity is $\vec{v}(t) = \langle \ln(t^2 + 1), 3\cos(t) \rangle$. What calculator command must be used to find this vector, and what is its 3-decimal numerical value?

19 **[Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion**

[AP Calculus BC Free-Response Questions (FRQs) · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

20 **[Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule**

Level 1 · FoundationsFind all values of c in the open interval $(-1, 2)$ that satisfy the Mean Value Theorem for $f(x) = x^3 - 2x^2 + 1$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

© ClipSAT — clipsat.org | All Rights Reserved

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which three conditions must be satisfied for a function f to be continuous at $x = c$?

a. $f(c)$ is defined, $\lim_{x \to c} f(x)$ exists, and $\lim_{x \to c} f(x) = f(c)$

b. $f'(c)$ exists, $f(c) > 0$, and $\lim_{x \to c} f(x) = 0$

c. $f(c)$ is continuous on (a, b), $f(c) = 0$, and $f'(c) \neq 0$

d. $\lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x)$ only

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Let $f(x) = \frac{x^2 - 9}{x - 3}$ for $x \neq 3$. What value must be assigned to $f(3)$ to make f continuous at $x = 3$?

a. 3

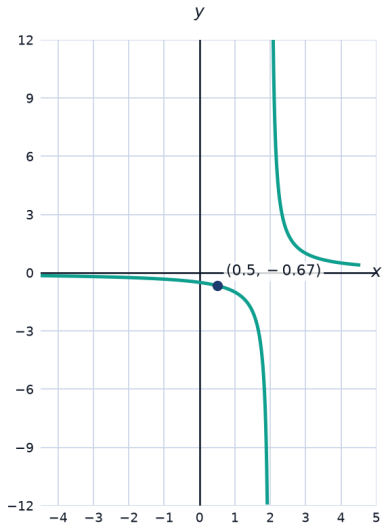
b. 6

c. 0

d. No value can make it continuous

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The function $g(x) = \frac{1}{x - 2}$ has what type of discontinuity at $x = 2$?



a. Infinite (essential) discontinuity

b. Removable discontinuity

c. Jump discontinuity

d. Oscillating discontinuity

© ClipSAT — clipsat.org | All Rights Reserved

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For what value of k is $f(x) = kx^2 + 1$ for $x \leq 2$, and $f(x) = 3x - 1$ for $x > 2$ continuous at $x = 2$?

a. 1

b. 2

c. 3

d. 0

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $g(x) = \frac{\sin(ax)}{x}$ for $x \neq 0$, and $g(0) = 4$, what value of a ensures continuity at $x = 0$?

a. 1

b. 2

c. 4

d. $\frac{1}{4}$

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The function $h(x) = x + 2$ for $x < 1$, and $h(x) = 2x + 1$ for $x \geq 1$ has what property at $x = 1$?

a. Continuous because both one-sided limits equal 3

b. Jump discontinuity of size 1

c. Infinite discontinuity

d. Undefined at $x = 1$

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What are the hypotheses of the Intermediate Value Theorem for a function f on interval $[a, b]$?

a. f must be continuous on the closed interval $[a, b]$

b. f must be differentiable on (a, b)

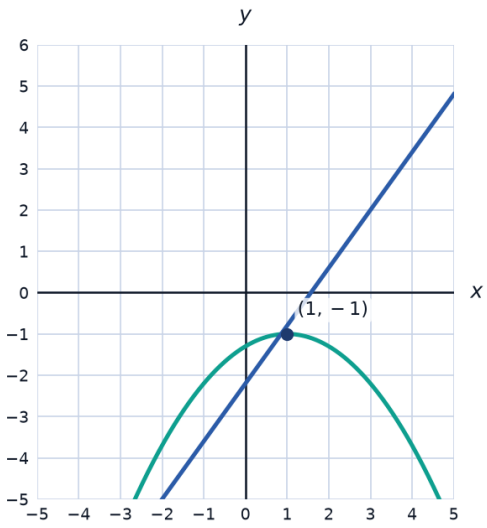
c. $f(a)$ and $f(b)$ must have opposite signs

d. f must be strictly increasing on $[a, b]$

© ClipSAT — clipsat.org | All Rights Reserved

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Let f be continuous on $[1, 5]$ with $f(1) = -3$ and $f(5) = 7$. Does the IVT guarantee a solution to $f(c) = 0$ for some $c \in (1, 5)$?



a. Yes, because 0 lies between -3 and 7

b. No, f must also be differentiable

c. No, f must be a polynomial

d. Only if $f'(x) > 0$ for all x

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

On which interval is the polynomial $P(x) = x^3 - 3x - 1$ guaranteed to have a real zero by the IVT?

a. $[1, 2]$

b. $[0, 1]$

c. $[-1, 0]$

d. $[2, 3]$

© ClipSAT — clipsat.org | All Rights Reserved

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What does the Extreme Value Theorem guarantee for a function f continuous on a closed interval $[a, b]$?

a. f attains both an absolute maximum and an absolute minimum on $[a, b]$

b. f has a horizontal tangent on (a, b)

c. $f'(c) = 0$ for some $c \in (a, b)$

d. f is invertible on $[a, b]$

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Why does the function $f(x) = \frac{1}{x}$ NOT attain an absolute maximum on $(0, 2]$?

a. The interval is open at 0 and f increases without bound as $x \to 0^+$

b. f is not differentiable on $(0, 2]$

c. f is bounded on $(0, 2]$

d. $f(2) = 0.5$ is the maximum

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the absolute maximum value of $f(x) = -x^2 + 4x + 1$ on the closed interval $[0, 3]$.

a. 5

b. 4

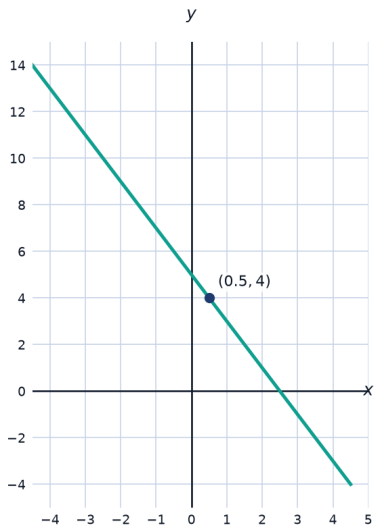
c. 1

d. 6

© ClipSAT — clipsat.org | All Rights Reserved

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The graph of a function $g(x)$ is shown above. The function $g(x)$ is defined by $g(x) = \frac{1}{3}x^3 - ax^2 + 2x + 1$, where a is a positive constant. The graph has a tangent line at $x = 1$ which is parallel to the line $y = -2x + 5$. Find the value of the constant a.



a. 1

b. 2

c. 3

d. 4

34 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Let $f(x)$ be a function such that $f'(x) = \cos(x^2)$ and $g(x)$ be a function defined by $g(x) = \sqrt{x}$. ; ; \textbf{(a)} Find the derivative of the composite function $h(x) = f(g(x))$. ; ; \textbf{(b)} If $f(1) = \pi$, find the value of $h'(4)$.

a. $\frac{\cos(4)}{2}$

b. $\frac{\pi\cos(4)}{2}$

c. $\frac{\cos(16)}{2}$

d. $\frac{\pi\cos(16)}{2}$

35 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

What is the value of $\lim_{h \to 0} \frac{(2+h)^4 - 16}{h}$?

a. 8

b. 16

c. 32

d. 64

© ClipSAT — clipsat.org | All Rights Reserved

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Consider the function $f(x) = x^2 - 3x$. If $g(x)$ is a function such that $h(x) = f(g(x))$, and at $x = 2$, $g(2) = 5$ and $g'(2) = 3$, find $h'(2)$.

a. 21

b. 30

c. 15

d. 6

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Let $f(x) = \sin(x^2)$. What is the derivative of $f(x)$?

a. $\cos(x^2)$

b. $2x\cos(x^2)$

c. $\sin(2x)$

d. $-2x\cos(x^2)$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

A particle moves along the x-axis. Its velocity is given by the function $v(t) = \sqrt{t^2 + 1}$ for $t \geq 0$. The position of the particle at time $t = 0$ is $x(0) = 5$. Consider the function $P(x)$ defined as the total displacement of the particle from $t = 0$ to $t = x$. That is, $P(x) = x(x) - x(0)$. Which of the following statements about $P(x)$ is true for $x > 0$?

a. $P(x)$ is always positive and increasing.

b. $P(x)$ is always negative and decreasing.

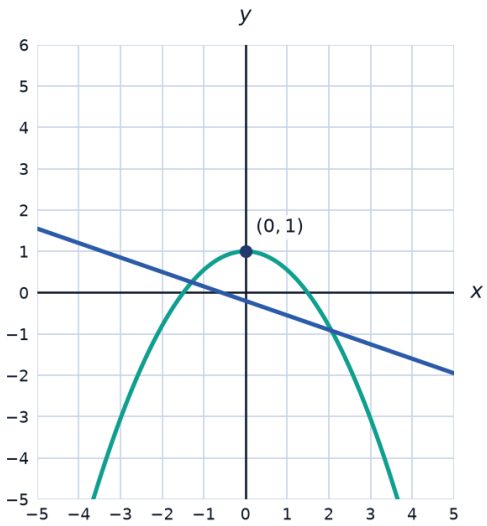
c. $P(x)$ is always positive and decreasing.

d. $P(x)$ is always negative and increasing.

© ClipSAT — clipsat.org | All Rights Reserved

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

A particle moves along a straight line so that its velocity at time $t \geq 0$ is modeled by $v(t) = 6t^2 - 18t + 12$. At what time t is the acceleration of the particle equal to 0?



a. $t = \frac{3}{2}$

b. $t = 1$

c. $t = 2$

d. $t = 3$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

Consider the function $f(x)$ whose graph is shown above. The graph of $f(x)$ consists of line segments and a quarter circle. Let $g(x) = \int_0^x f(t)\,dt$.

a. g'(2) = 2

b. g'(2) = 0

c. g'(5) = -2

d. g'(5) = 0

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

A particle's velocity is given by the function $v(t) = 5$ for $0 \leq t \leq 4$. Find the total displacement of the particle over this time interval.

a. 10

b. 15

c. 20

d. 25

© ClipSAT — clipsat.org | All Rights Reserved

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

If $f(x) = x^2 + 3$ is increasing on $[1, 6]$, a left Riemann sum with equal subintervals will:

a. equal the true area exactly

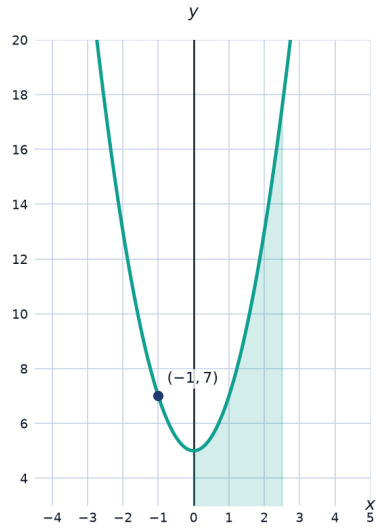
b. underestimate the true area

c. overestimate the true area

d. cannot be determined

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

If $f(x) = 2x^2 + 5$ is increasing on $[2, 5]$, a right Riemann sum with equal subintervals will:



a. equal the true area exactly

b. cannot be determined

c. underestimate the true area

d. overestimate the true area

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Using a right Riemann sum with 4 subintervals of equal width on $[0, 2]$, approximate $\int_0^2 (x^2 + x - 2)\,dx$.

a. $\frac{21}{4}$

b. $-\frac{3}{4}$

c. $-\frac{15}{4}$

d. $\frac{9}{4}$

© ClipSAT — clipsat.org | All Rights Reserved

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Using the Trapezoidal Rule with 2 subintervals on $[0, 6]$, approximate $\int_0^6 (2x^2 - 3x - 1)\,dx$.

a. 183

b. 21

c. 104

d. 102

© ClipSAT — clipsat.org | All Rights Reserved